Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/25/2007
PI Name:	Cabrera, Marco E. Ph.D.		
Project Title:	Time Course of Metabolic Adaptations During Loading and Unloading		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:	NIH/National Institute of Biomedical Imaging and Bioengineering	**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106-6011	**Congressional District:**	11
Comments:	NOTE: PI passed away in early February 2009.		
Project Type:	Ground	**Solicitation / Funding Source:**	2004 NSF 04-607 Interagency Solicitation Multi-Scale Modeling
Start Date:	06/12/2006	**End Date:**	03/31/2009
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dash, Ranjan K. (Case Western Reserve University) Silva , Fatima (Case Western Reserve University) Previs , Stephen (Case Western Reserve University) Puchowicz , Michelle (Case Western Reserve University) Bederman, Ilya (Case Western Reserve University)		
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Performance Goal No.:			
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Task Description:	The alterations in skeletal muscle structure and function induced by space travel or endurance training are the result of a chronic mechanical stimulus (or lack of it) a series of metabolic interactions spanning from the cellular to the organism level. Reduced activation of the weight bearing muscles alters gene expression of myosin heavy chain isoforms to meet the functional demands of weightlessness and results in muscle atrophy, reduced capacity to process fatty acids, and reduced muscle endurance. On the other hand, endurance training induces an up-regulation of oxidative enzymes and results in increased fatty acid oxidation capacity and muscle endurance. The few human and animal studies comparing skeletal muscle structure and function before and after unloading suggest that the adaptations are time and biological-level dependent. While some parameters may be altered within a week (muscle mass, insulin resistance), others may take longer to show changes (work efficiency, VO2max). Similar conclusions may be drawn from endurance training studies. Thus, to establish countermeasures aimed at restoring astronauts' work capacity, it is critical to determine: (1) the time course of muscle atrophy, (2) the bioenergetic, metabolic, and fuel-processing factors that contribute to the reductions in endurance associated with muscle atrophy, and (3) the effects of skeletal muscle atrophy on whole-body metabolism. To answer these questions a multi-scale model of energy metabolism -linking cellular to whole organism processes- is needed in combination with appropriate cellular to whole-body measurements collected during the period of adaptation. The objective of this proposal is to identify the time course of the adaptations to loading and unloading and to integrate them using a multiscale model of skeletal muscle and whole body metabolism, in order to predict the system integrated response after periods of space travel or endurance training. Accordingly, (1) A team of two computational modelers with expertise on scales ranging from cellular to whole body and 3 experimentalists with expertise in complementary methodologies in rodents (cell biochemistry [MAP], tracers and tissue metabolism [SFP], cardiorespiratory control [FTS] and whole body physiology [FTS, SFP]) has been formed to reach the proposal's objective. (2) A novel top-down approach will be applied to model whole-body metabolism and to bridge organism to cellular level. Thus, a minimal, but comprehensive, systems model of metabolism will be developed which includes: (a) brain, heart, liver, adipose tissue, and skeletal muscle, (b) a cardiorespiratory controller, (c) pancreas and liver, and (d) the cardiovascular system. Modeling transport processes across membranes will constitute an essential feature of the integration process linking the various scales. For model validation, molecular and cellular measurements will be used to assessed tissue and whole body metabolism. To maintain intact regulatory mechanisms, a novel methodological approach with emphasis on the use of in vivo measurements will be applied, specifically, proteomics and metabolomics using state-of-the-art stable isotope techniques. (3) This multi-scale model of metabolism will enable testing of physiologically-feasible time profiles of the adaptations to loading/unloading in every organ/tissue, which would otherwise require collection of hundreds of samples in thousands of animals. Specifically, this model can be used to predict tissue metabolism (e.g., fuel efficiency, buffering capacity, and muscle endurance) as well as whole body metabolism (e.g., insulin resistance, lactate acidosis, work capacity). The long-term goal of this project is to predict astronaut's energy metabolism and performance during altered loading states using a multi-level model of the human metabolic and energetic system in order to develop countermeasures. We envision that this computational model of human bioenergetics would become an essential component of the "Digital Astronaut". Thus, the specific aims of this proposal are: 1) To identify and model the time course of the structural, metabolic and functional adaptations to loading and unloading. 2) To develop and validate a multi-level (cell-muscle-whole body) computational model skeletal muscle metabolism in the context of the whole body. 3) To predict the integrated response of muscle fibers, whole skeletal muscle, and whole body during periods of loading and unloading, in particular astronaut's muscle energetics and function, as well as his/her whole body metabolism and performance. Involved with the Associate Director of NIH Center for Modeling Integrated Metabolic Systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Using a multiscale model of skeletal muscle and whole body metabolism will enable prediction of the system integrated response after periods of space travel or endurance training.
Task Progress:	In the first year of funding, our goals were to (i) set up loading (exercise) and unloading (hindlimb suspension) systems, (ii) initiate time-course HSU studies, (iii) set up analytical protocols to obtain muscle and whole-body parameters (enzyme activities and metabolite concentrations) for multi-scale modeling under loading/unloading conditions, and (iv) initiate incorporation of biological outcome parameters into modeling algorithms. In this report, we demonstrate that during first 7-9 months of funding, our group successfully developed HSU and exercise systems and provided initial metabolic parameters which agree with most common literature findings in the field of loading/unloading research. In parallel, we are developing multi-scale modeling system using biological parameters obtained during our initial animal studies. Our plan for the next 6 months includes (i) executing a longer 4-week hindlimb study to obtain metabolic parameters; (ii) executing a 2- and a 4-week hindlimb studies using doubly-labeled water to measure whole-body energy balance and metabolic fluxes in protein and fat metabolism; and (iii) execute a pilot HSU experiment, where dietary conditions are varied between regular low-fat, high-fat, and carbohydrate-free diets; and (iv) model the time-course of metabolic changes that occur as the result of HSU and/or dietary intervention.
Bibliography Type:	Description: (Last Updated: 08/15/2018)
Abstracts for Journals and Proceedings	Lai N, Camesasca M, Saidel GM, Dash RK, Cabrera ME. "Linking Pulmonary Oxygen Uptake, Muscle Oxygen Utilization and Cellular Metabolism during Exercise." Biomedical Engineering Society Annual Meeting, Chicago, IL, October 2006. Ann Biomed Eng. 2007 Mar 23 (In Press). , Mar-2007

Articles in Peer-reviewed Journals	Lai N, Camesasca M, Saidel GM, Dash RK, Cabrera ME. "Linking Pulmonary Oxygen Uptake, Muscle Oxygen Utilization and Cellular Metabolism during Exercise." Ann Biomed Eng. 2007 Mar 23; [Epub ahead of print] PMID: 17380394 , Mar-2007
Articles in Peer-reviewed Journals	Lai N, Dash RK, Nasca MM, Saidel GM, Cabrera ME. "Relating pulmonary oxygen uptake to muscle oxygen consumption at exercise onset: in vivo and in silico studies." Eur J Appl Physiol. 2006 Jul;97(4):380-94. PMID: 16636861 , Jul-2006
Articles in Peer-reviewed Journals	Dash RK, Dibella JA 2nd, Cabrera ME. "A computational model of skeletal muscle metabolism linking cellular adaptations induced by altered loading states to metabolic responses during exercise." Biomed Eng Online. 2007 Apr 20;6(1):14 [Epub ahead of print] PMID: 17448235 , Apr-2007